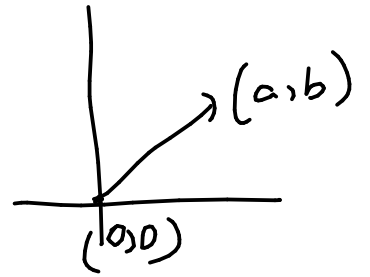


2D vector : in XY plane $\begin{bmatrix} a \\ b \end{bmatrix}$ (or) (a,b)

denotes a vector from origin to point (a,b)



in our discussion we talk about direction vectors mostly. not

position vectors. and direction starting from origin only. not mentioned everywhere but when i say vector (a,b) its a direction vector from $(0,0)$ to (a,b)

Since i am saying Direction Vector do not care about magnitude for this.

so (a,b) and $(2a,2b)$, $(-9a,-9b)$

are on same line

$\downarrow$
reversed direction
but on same
line.

3D all same, we talk about direction only from origin $(0,0,0)$ to (x,y,z)

Transformations in 2D

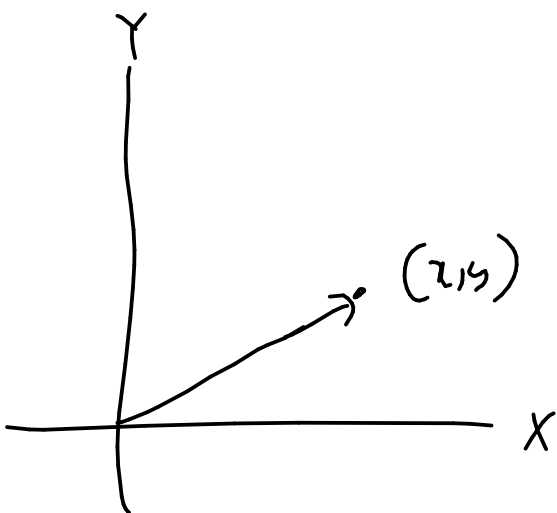
First we'll see elementary transformations,
later we'll go to compound transformations,
multiple transformations applied in order.

① rotation in 2D

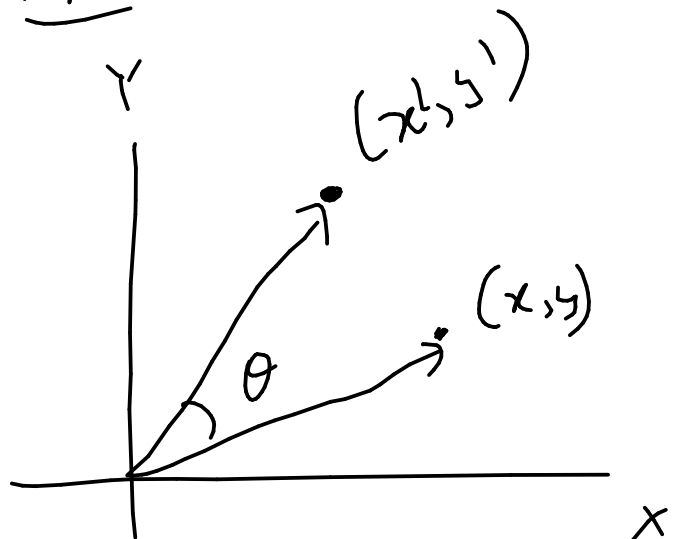
imagine a XY plane and a point (x, y) on it.

→ Now keep the XY axes fixed and
rotate all points by angle θ anticlockwise
with respect to origin. (only rotating all
points, not axes)

originally



later



The point (x, y) will shift to (x', y') .

(x', y') with respect to old coordinate system only.

we know already, $\rightarrow$ (im not writing proof here)

$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

So given any point (x, y) in old system after doing this transformation we get new coordinates (x', y') by using this 2 eq's.

Writing this in matrix form

Try to understand this completely first

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Vector for new point

Transformation matrix for this operation.

old point in a vector

2×2 , 2×1 matrix multiplication.

ex: assume $\theta = 90^\circ$, old point = $(5, 0)$. where will it shift?

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} \cos 90 & -\sin 90 \\ \sin 90 & \cos 90 \end{bmatrix} \begin{pmatrix} 5 \\ 0 \end{pmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} 5 \\ 0 \end{pmatrix}$$
$$= \begin{bmatrix} 0.5 + -1.0 \\ 1.5 + 0.0 \end{bmatrix} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

$(5, 0)$ lies on x axis,
after shifting by 90°
anticlock it jumps to
y axis at $(0, 5)$

now try to see it
visually

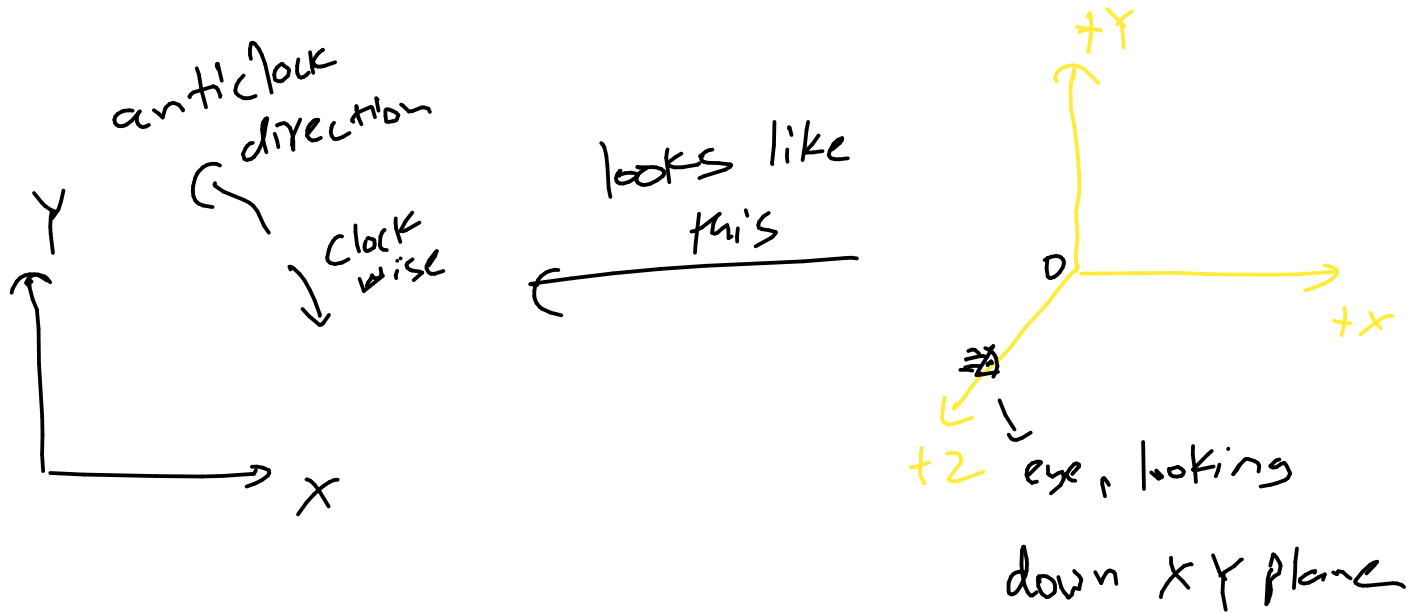
Note: I said keep axes fixed and rotate^{all}
points by θ anticlockwise, and look at
new position of the points from old axes
only $\rightarrow$ it will be (x', y') .

(OR) keep the points there only and rotate
the axes by θ in clockwise direction. w.r.t

new axes system, the original point (x, y)

will appear at (x', y') $\rightarrow$ same (x', y')

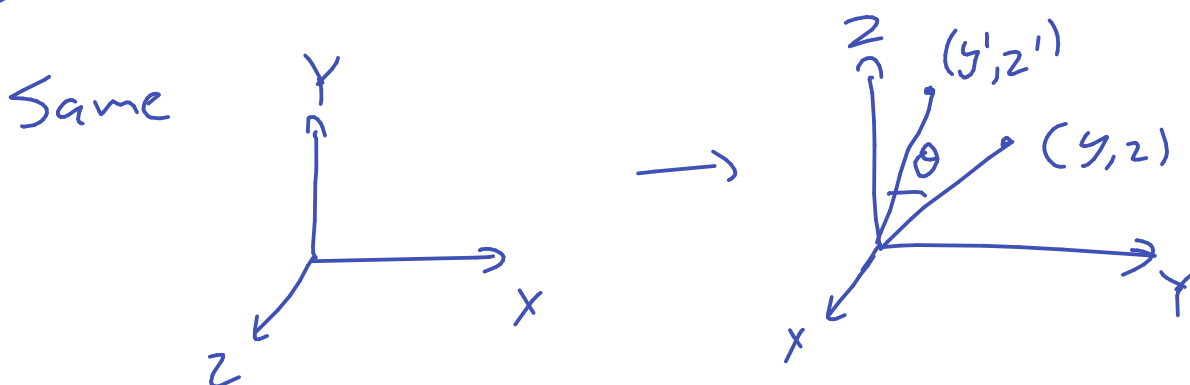
Note-2 :- we are doing this in XY plane by looking downward to origin from +z direction. and rotating by ' θ ' (anticlockwise)



Variations in rotation operation :-

① what if we rotate by ' θ ' but in clockwise direction $\rightarrow$ just substitute ' $-\theta$ ' in the c-s matrix.
 c s
 s c

② what if we do this in YZ plane.

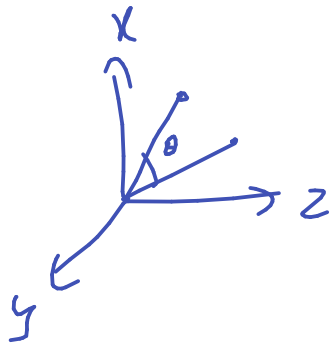


$$y' = y \cos \theta - z \sin \theta$$

$$z' = y \sin \theta + z \cos \theta$$

$$\Rightarrow \begin{bmatrix} y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix}$$

Same for zx .

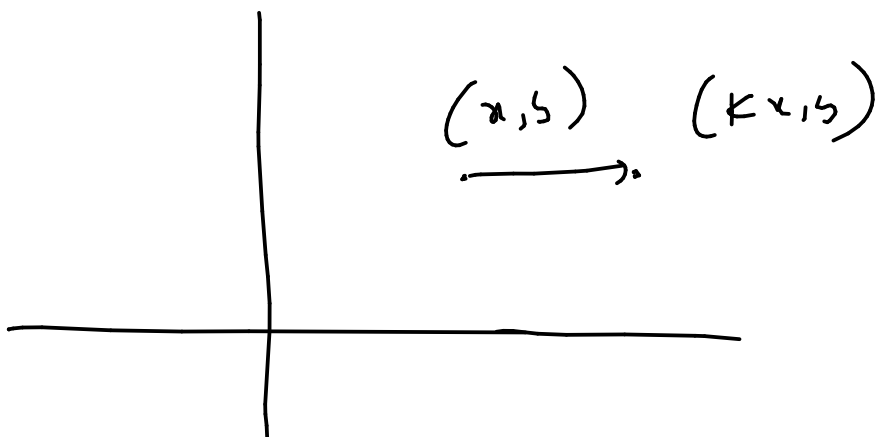


$$\begin{bmatrix} z' \\ x' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} z \\ x \end{bmatrix}$$

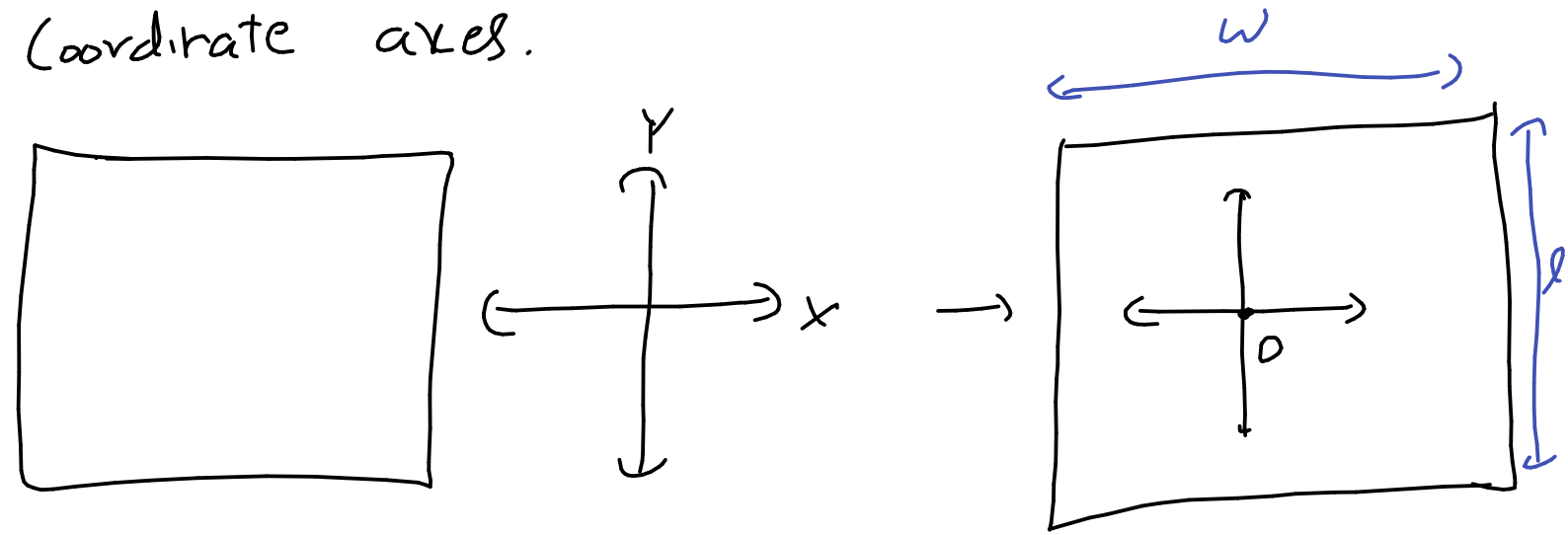
② Scaling operation (ignore the 30cm scale thing i said in class)

Scaling is just, given a point (x, y) , if you are scaling all points along x direction by K . then all (x, y) points become (Kx, y) .

Visualize as,



its like you just have a paper which is your XY plane. And you put 2 perpendicular sticks on top of paper and you call them coordinate axes.



→ and you gave every point on paper some coordinates based on origin which is just the intersection point of them sticks.

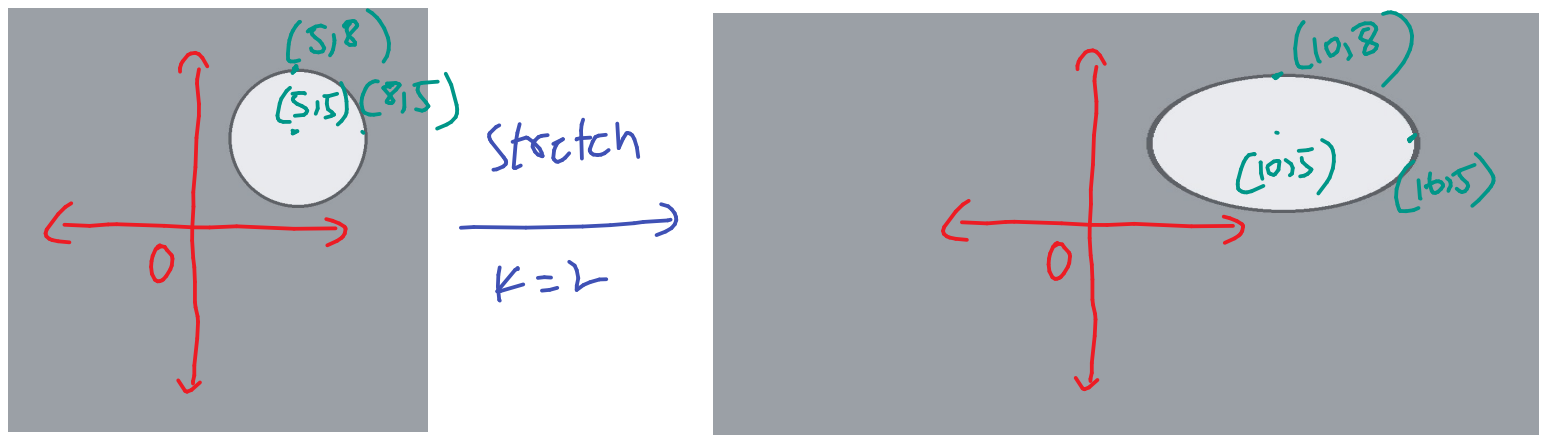
Now.....;

Put the sticks aside, stretch the paper in X Direction so that its width becomes K times but length is unchanged.

→ and put the sticks back on paper.

→ Make sure new origin lies on same origin as before.

If you look at rest of all points wrt the sticks as axes. every point which was at (x, y) previously would now be at (kx, y) now, (wrt same sticks as axes)



the images are identical copies, one is stretched. the axes drawing also identical, (copy pasted).

So, you should be able to visualize the scaling operation. If we scale every thing along x direction by k . every point (x, y) becomes (kx, y) .

→ you can apply scaling in y direction also along with this.

if in x direction you are scaling by factor ' a ' and in y direction by ' b '. then wrt unchanged axes. old point at (x, y) moves to (ax, by) .

Transformation matrix for scaling is

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

if a, b both $= 1$

we aren't doing anything

then $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is I , $x' = x$
 $y' = y$

Note: a special case of scaling is flipping the x axis in reverse direction.

$$k = -1$$

all (x, y) becomes $(-x, y)$ in new system.

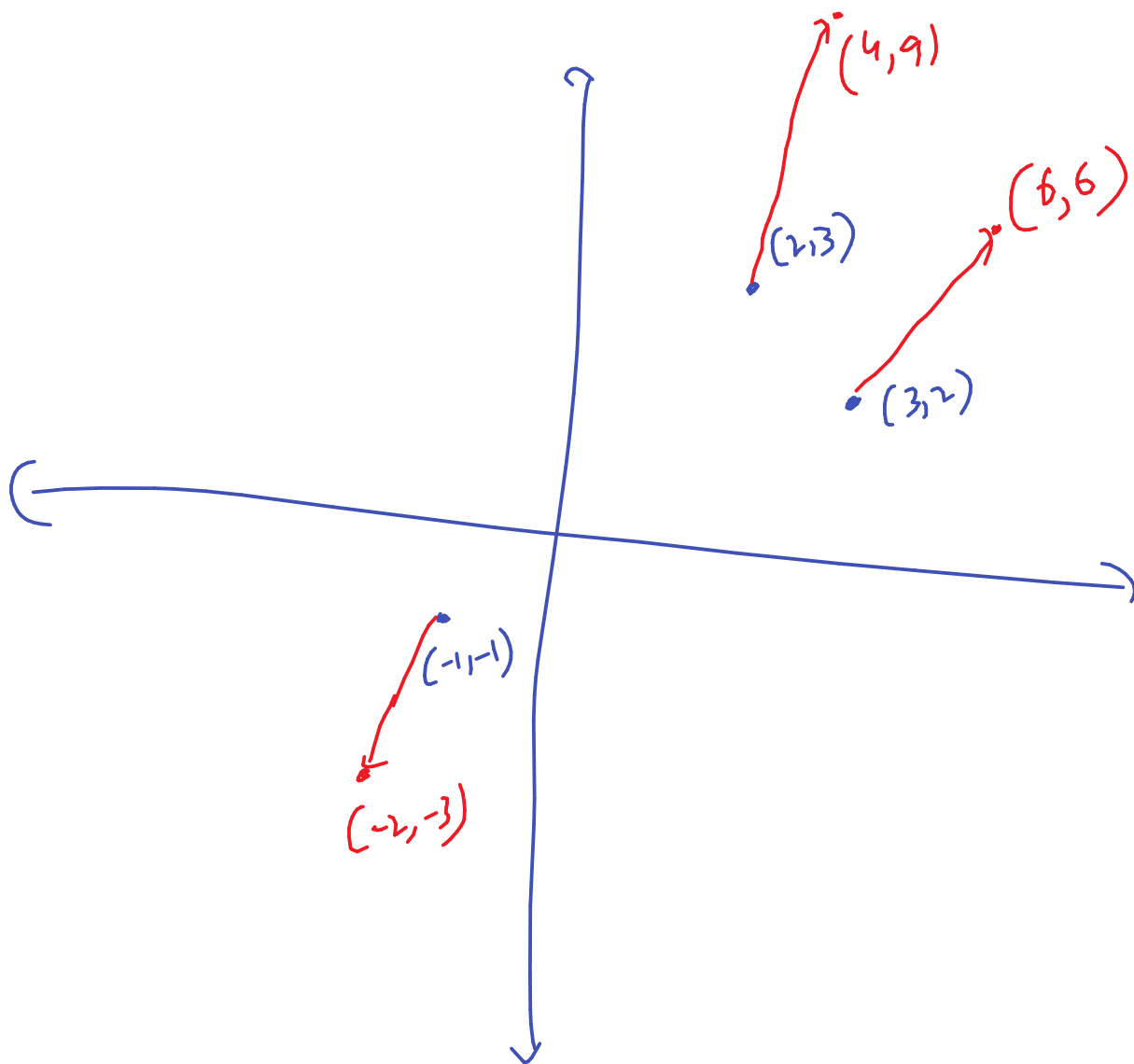
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

ex: if we are scaling along x direction by $K_1=2$, scaling in y by $K_2=3$. where will the points $(2,3)$, $(3,2)$, $(-1,-1)$ go to

$$(2,3) \rightarrow (2 \cdot 2, 3 \cdot 3) = (4,9)$$

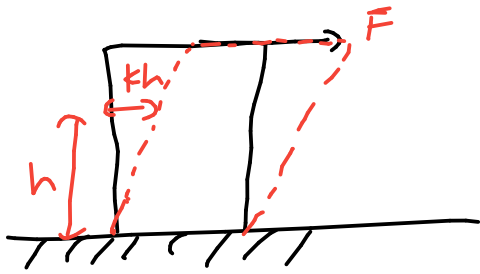
$$(3,2) \rightarrow (2 \cdot 3, 3 \cdot 2) = (6,6)$$

$$(-1,-1) \rightarrow (2 \cdot -1, 3 \cdot -1) = (-2,-3)$$



③ Shearing operation in 2D:

imagine a block sitting on a surface.
and its glued to ground.



→ Now force is applied
on top.

→ the block shifts
little bit.

→ at a height 'h'. all points in the block
shift by some $k \cdot h$ amount horizontally.

→ This is caused by shear stress by force F .

→ we call ' k ' as shear constant here.

$$k \propto F.$$

In our content,

all points (x, y) shift to $(x + ky, y)$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

→ For all points having $y=0$, they don't change anything

→ Most shift happens to points at top.

Shear along Y axis

if shear happens along Y with $\text{const} = k$

then $(x, y) \rightarrow (x, y + kx)$

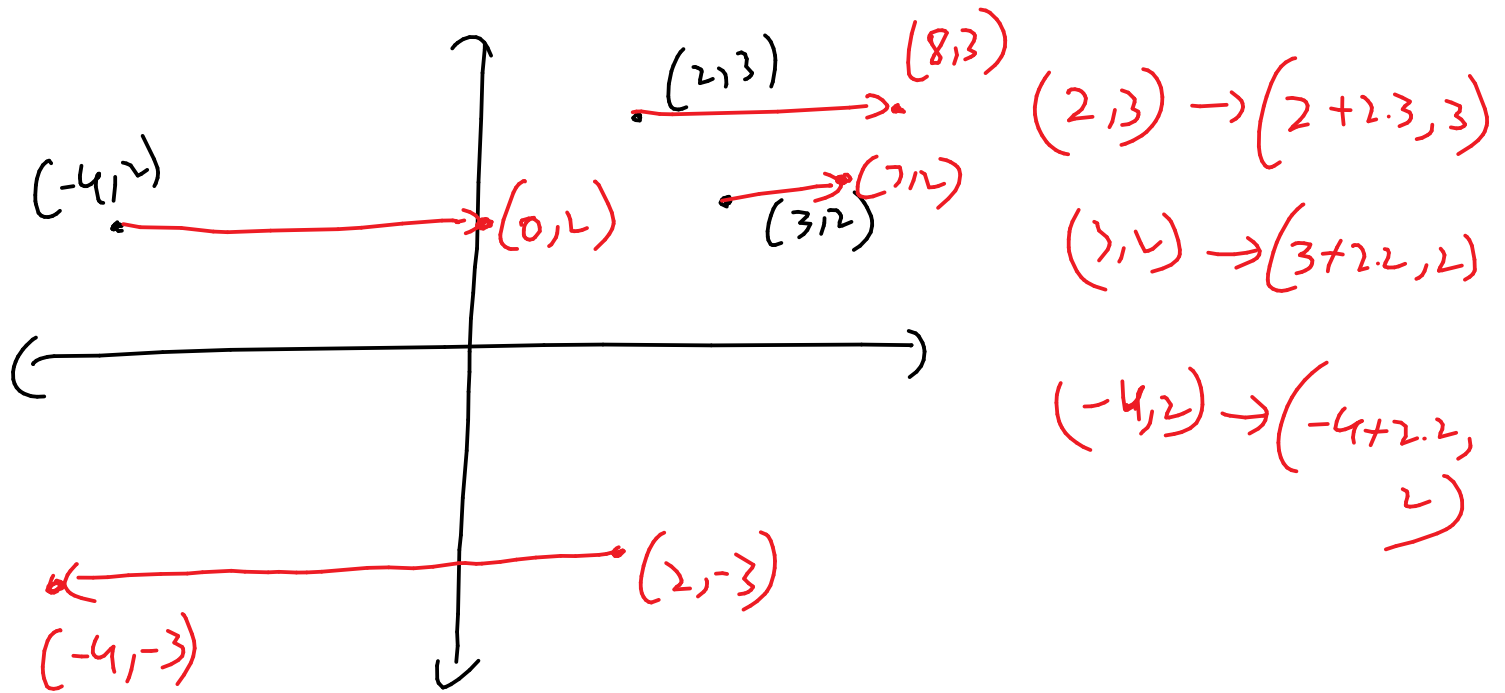
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

In general if shear applied on both directions simultaneously with constants k_x, k_y .

then $(x', y') \rightarrow (x + k_x y, y + k_y x)$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x + k_x y \\ y + k_y x \end{bmatrix} = \begin{bmatrix} 1 & k_x \\ k_y & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

ex:- if shearing is applied along X
with $k = 2$. where will the points $(2, 3)$,
 $(3, 2)$, $(-4, 2)$ shift to?



and for point $(2, -3)$? $\rightarrow (2 + 2 \cdot -3, -3)$
 $= (-4, -3)$

Q)

If there is a point $(a, -4)$ in above example
such that, after shearing it lies on
Y axis. what is value of a ?

after shearing $x' = 0 \Rightarrow a + 2(-4) = 0$
 $a = 8$

Till now we only saw elementary transformations we can do any complex transformation based on these small transformations applied one by one. we'll see that shortly.

→ We haven't seen shifting of origin also till now. we'll see that too.

→ Rotation, Scaling, Shearing in 3D

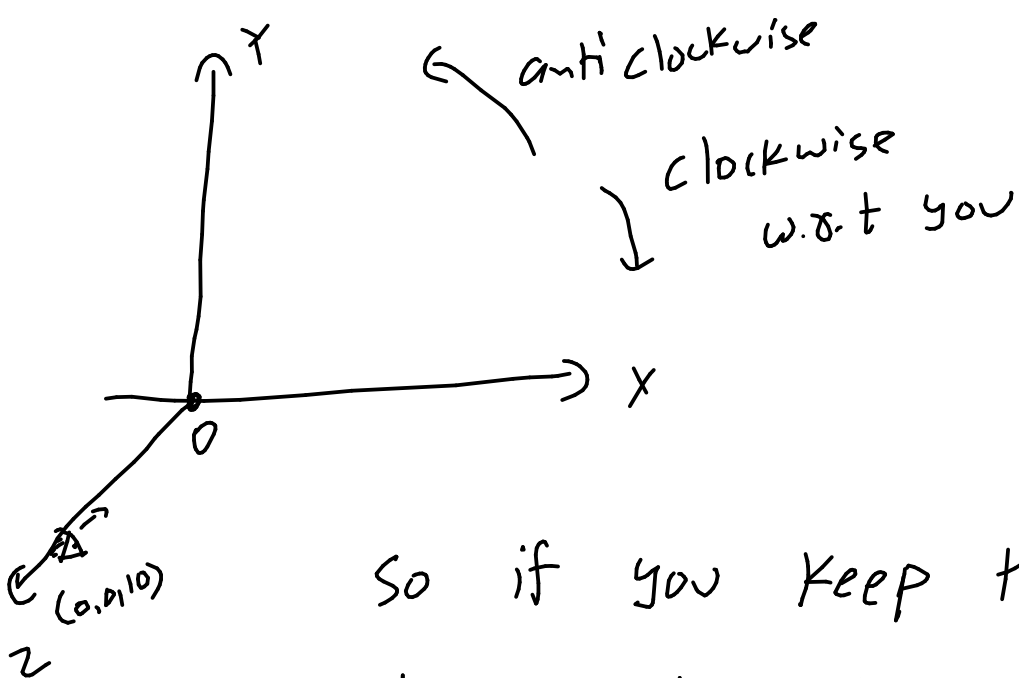
Rotation in 3D

Rotation is always done around a single axis. For now we'll talk about Transformation matrix for rotation around X axis, Y axis, Z axis only individually.

→ rotation around an arbitrary line in 3D we'll see later.

3D rotation around Z axis.

imagine you are at a point on Z axis at $(0, 0, 20)$. you are looking towards origin. you'll see this



So if you keep the axes fixed and rotated the entire 3D space around Z-axis anticlockwise by θ

then every point (x, y, z) will become (x', y', z')

where $x' = x \cos \theta - y \sin \theta$

$$y' = x \sin \theta + y \cos \theta$$

$$z' = z \rightarrow \left(\text{Since we are rotating around } z\text{-axis, this won't change} \right)$$

So transformation matrix will be,

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Similarly, if you do it wrt x axis then you can write,

$$\begin{bmatrix} y' \\ z' \\ x' \end{bmatrix} = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y \\ z \\ x \end{bmatrix}$$

but we use the term transformation matrix strictly

for (x, y, z) in order so.

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

↓

Transformation matrix for rotating around x axis

Note :- For rotation around an arbitrary line. we will do the following (more about this later)

→ we have to do 2 rotations

to bring our X axis along this arbitrary direction

→ then rotate along new X axis.

→ rotate X axis to original position.

(detailed we'll see later)

Scaling in 3D

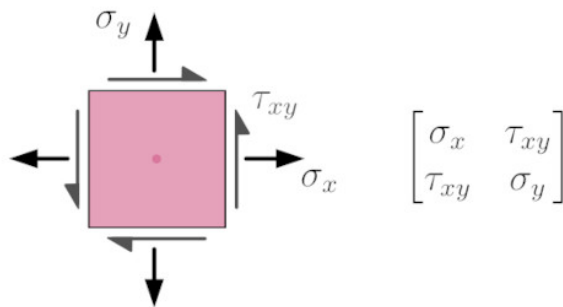
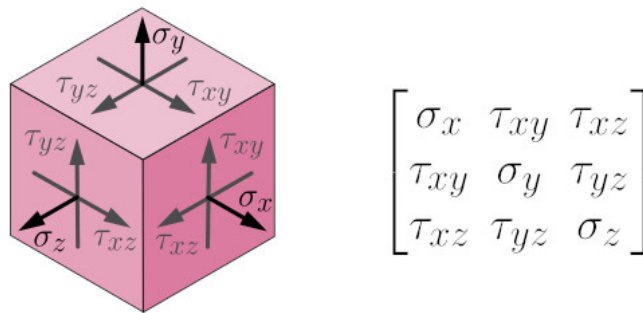
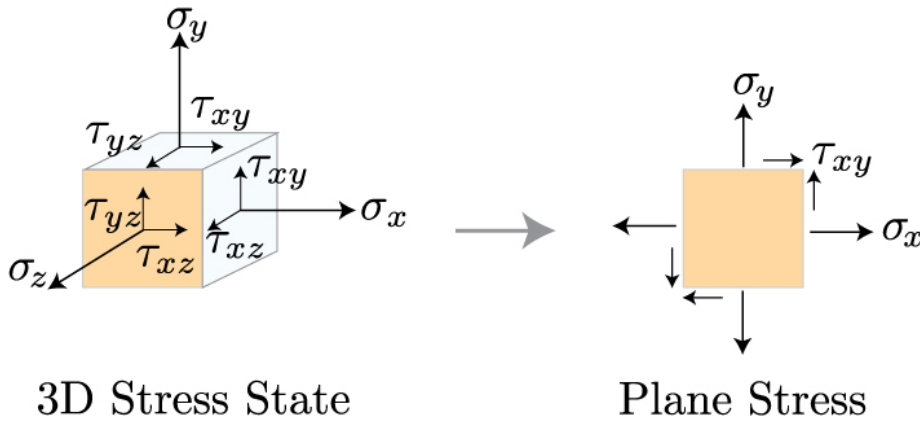
→ Scaling along X by factor S_x , along y by factor S_y , z by S_z

original points $(x, y, z) \rightarrow (S_x \cdot x, S_y \cdot y, S_z \cdot z)$

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & S_z \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

easy. but
scaling in arbitrary
direction is not
simple. we'll see
later.

Shearing in 3D $\rightarrow$ Shear forces in 3D are complicated. They have components in all planes and all directions.



So we will skip shearing in 3D.
(not important at our level).

Till now we saw different transformations.
all elementary transformations only. now lets
see how to do complex transformations
using these elementary transformations.

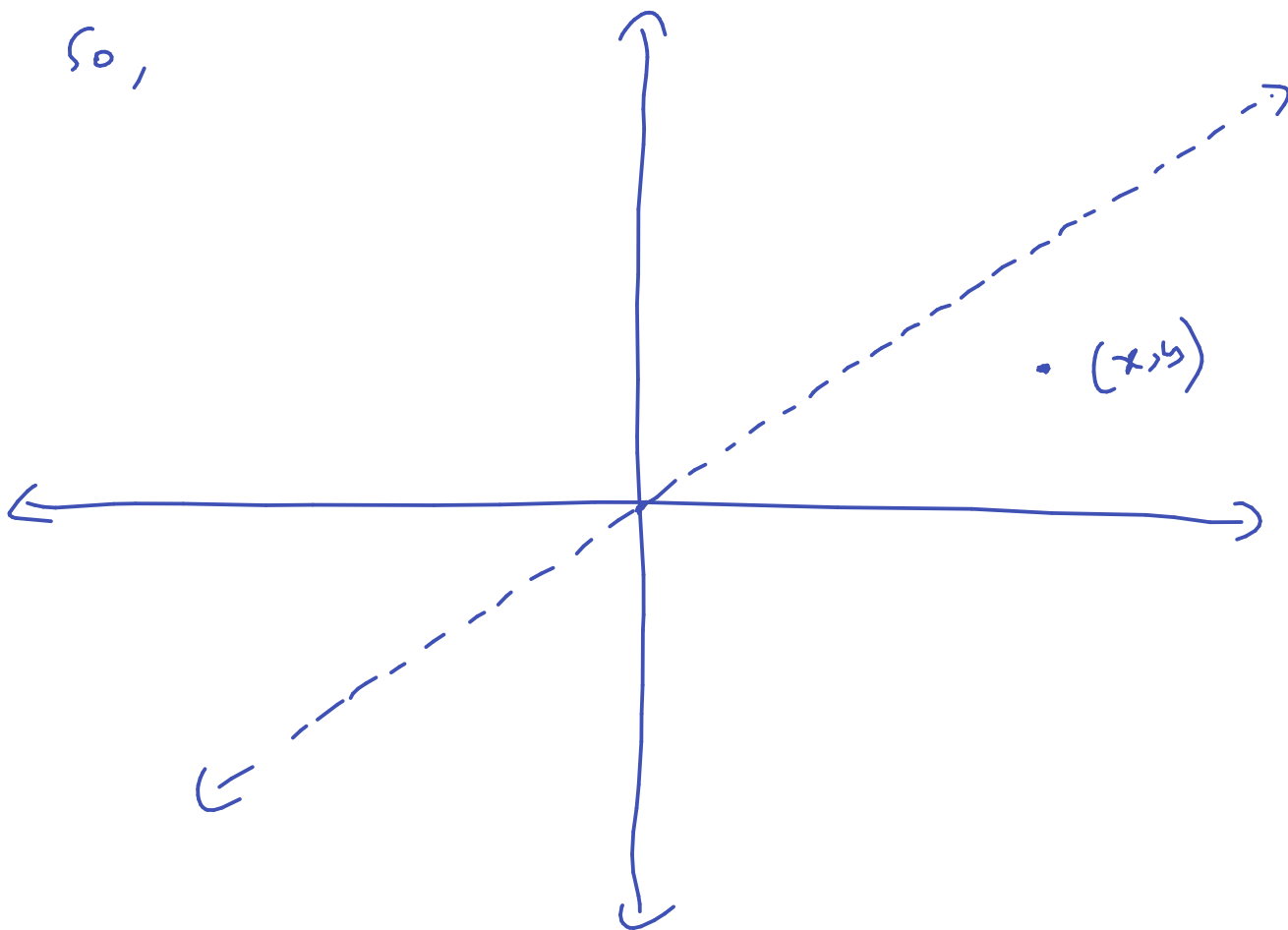
Series of transformations :-

If you want to do multiple transformations
in an order. The transformation matrix
for whole thing will be the product of
transformation matrices of all individual
operations in order. will see with examples.

① Scaling in 2D . but in $y=x$ direction.
with origin as midpoint.

we want to do scaling but in $y=x$ direction
by some factor 'k'.

So,



Previously we did scaling in x direction with origin as mid point, meaning we stretched the xy plane in x direction keeping origin unchanged.

→ Now we want to stretch in $y=x$ direction instead.

→ We will use more than one elementary transformation to achieve this.

Idea :-

① Keep the axes fixed. Rotate the plane Clockwise by 45° so that $y=x$ line aligns with X-axis. The points will also rotate.

then (x, y) becomes (x', y') now where

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad \text{--- ①}$$

↙
 (x', y') are coordinates of Point (x, y) after rotating.

② now $y=x$ line is along X-axis so we can do scaling along X.

$$\begin{bmatrix} x'' \\ y' \end{bmatrix} = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} \quad \text{--- ②}$$

③ now after scaling is over, we need to rotate back all points by θ anticlock again,

So final (X, Y) will be

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad \text{--- (3)}$$

→ We just rotated X axis to align with $y=x$ then scaled along X , now bringing X axis back to original place. To perform our transformation, we had to do 3 elementary transformations.

So to get (X, Y) from (x, y)

Substitute (1) in (2) and (2) in (3), we'll get

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} C & -S \\ S & C \end{bmatrix} \begin{bmatrix} K & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} C & S \\ -S & C \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\Downarrow$$
$$\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} K \cos^2\theta + \sin^2\theta & \sin\theta (\cos\theta (K-1)) \\ \sin\theta (\cos\theta (K-1)) & K \sin^2\theta + \cos^2\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

So that is the transformation matrix we got for our operation. let's see more about it

let's see if $k=1$, $k=1$ means no scaling,
Same as before

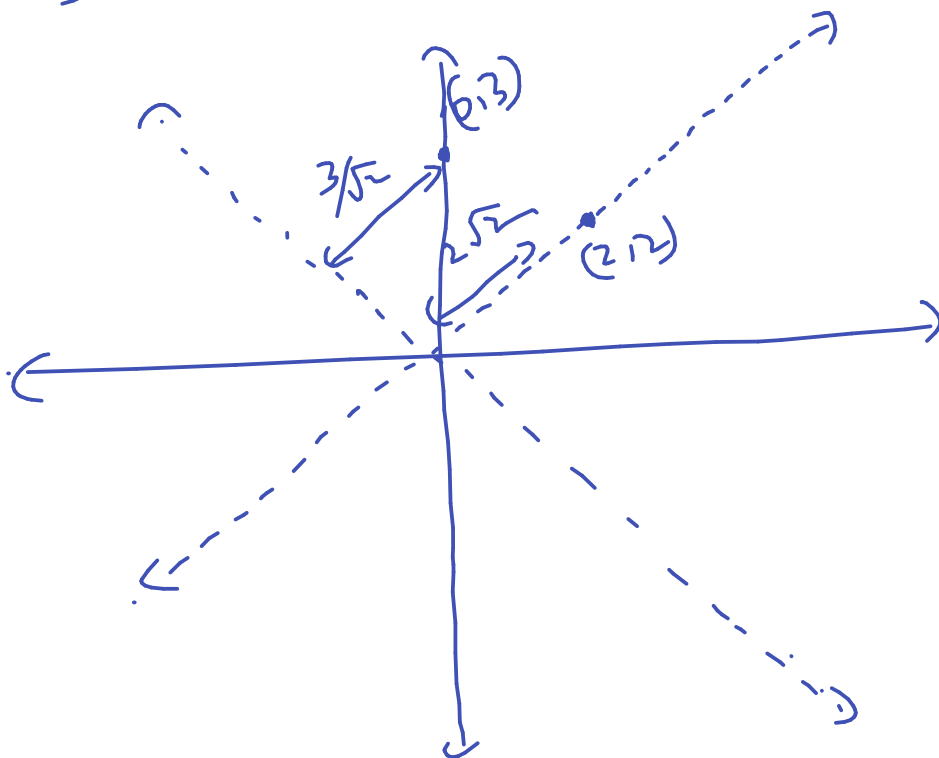
So if we put $k=1$

the matrix becomes $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ which is correct.

(Scaling along $y=x$, so $\theta=45^\circ$)

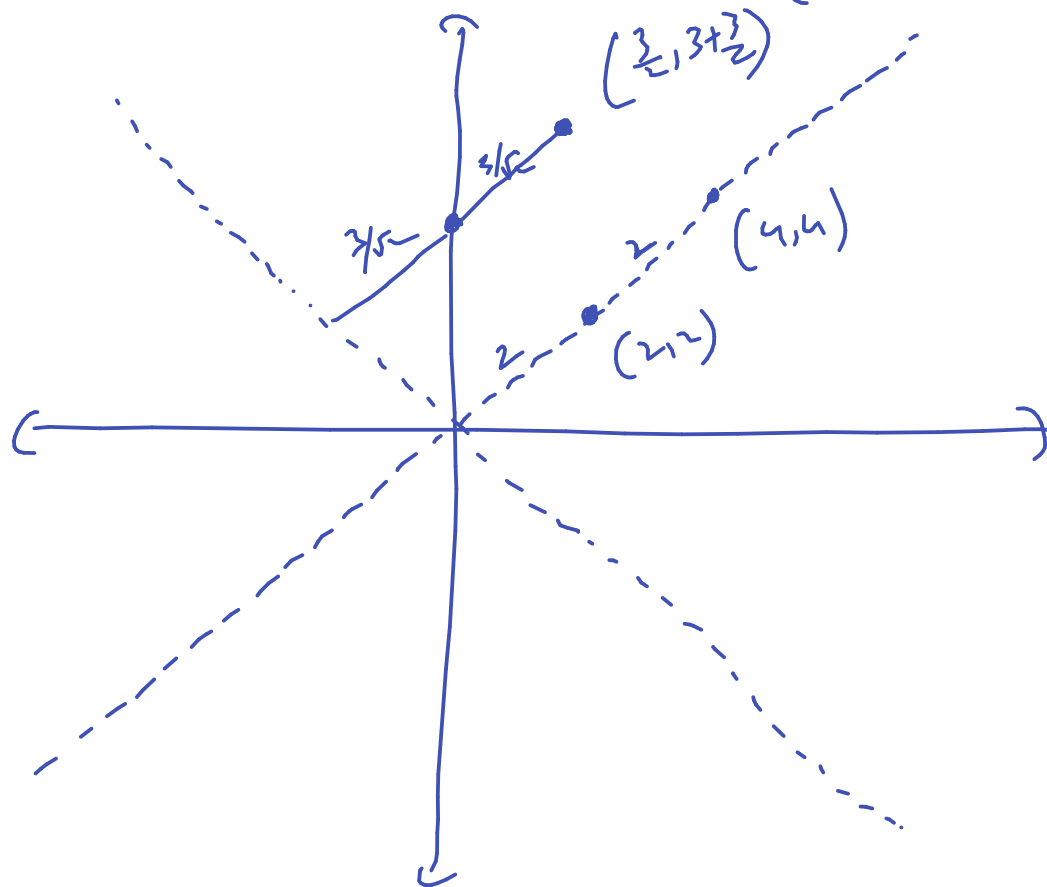
ex:- Consider the points $(2,2)$, $(0,3)$ and

$k=2$. first on diagram where will they
So to. then we'll apply matrix and verify
they're going to same point or not.



Since we're stretching along $(y=x)$ first find out distance of Point from $y=-x$ line.

So double that distance (since $k=2$)



$$(0, 3) \rightarrow \left(\frac{3}{2}, \frac{9}{2}\right), \quad (2, 2) \rightarrow (4, 4)$$

Now Verify if these are correct using matrix. for $k=2$, $\theta=45^\circ$ our matrix will be

$$\begin{bmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{bmatrix} \rightarrow \left(\begin{array}{l} \text{we saw this example} \\ \text{in slides, i said} \\ \text{it maybe shearing} \\ \text{but its} \\ \text{Scaling actually} \end{array} \right)$$

$$\begin{bmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 3/2 \\ 9/2 \end{bmatrix} \checkmark$$

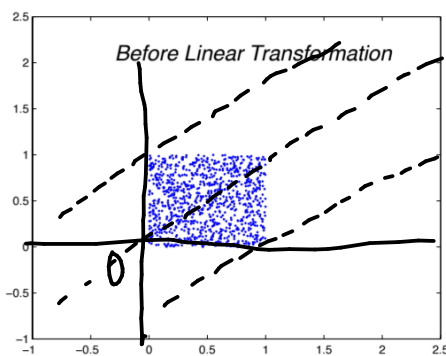
$$\begin{bmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} \checkmark$$

So our TM
is correct

■ Lets consider another example

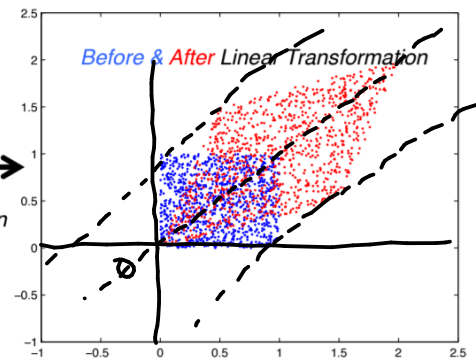
- This time non-zero diagonal elements

$$A = \begin{pmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{pmatrix}$$



$$A = \begin{pmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{pmatrix}$$

Linear Transformation



Scaled along $y=x$ by $k=2$

Main idea is that when we do a series of elementary transformations, the TM of whole transformation is equal to product of TMs of individual matrices.

if our whole transformation has series of transformations $TM_1, TM_2, \dots, TM_n$.

$$\text{then } \begin{bmatrix} x' \\ y' \end{bmatrix} = TM_1 \cdot \begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} x'' \\ y'' \end{bmatrix} = TM_2 \cdot \begin{bmatrix} x' \\ y' \end{bmatrix}, \dots$$

$$\dots \begin{bmatrix} x^{(n)} \\ y^{(n)} \end{bmatrix} = TM_n \begin{bmatrix} x^{(n-1)} \\ y^{(n-1)} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x^{(n)} \\ y^{(n)} \end{bmatrix} = \underbrace{TM_n \cdot TM_{n-1} \cdot TM_{n-2} \cdot \dots \cdot TM_1}_{\text{}} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

More examples

① Shearing along arbitrary direction in 2D.

↳ (i) rotate x axis till it aligns with that arbitrary direction, (TM_1)

(ii) apply shear transformation wrt new x-axis (TM_2)

(iii) rotate back x axis to original position. (TM_3)

→ So total TM will be $TM_3 \cdot TM_2 \cdot TM_1$,
for this transformation

② Consider the following problem of. In 2D we want to find $\perp^{\text{th}}$ projection of every point (x, y) on some line $(y = mx)$.

→ Naive way is what we saw in class, but we can do series of some transformations to achieve that.

↳ next page we'll see that

↳ 1st transformation :- rotate x axis to
(i) align with $y=mx$ line.

(ii) 2nd TM :- do scaling of every point along
y axis with $k=0$, basically collapsing
all points on to x axis. (new x axis).

(iii) 3rd TM :- rotate x axis back to original
position.

$$\text{total TM} = TM_3 \cdot TM_2 \cdot TM_1$$

$$= \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} c & s \\ -s & c \end{bmatrix}$$

where $s = \frac{m}{\sqrt{1+m^2}}$, $c = \frac{1}{\sqrt{1+m^2}}$ because $\tan \theta = m$

will get $\begin{bmatrix} \frac{1}{1+m^2} & \frac{m}{1+m^2} \\ \frac{m}{1+m^2} & \frac{m^2}{1+m^2} \end{bmatrix}$ we got
Same in class

③ in 3D, Consider a line passing through origin and you want to rotate all points by ' θ ' anticlock w.r.t this line.

idea:- first somehow rotate your axes in such a way that your z axis lies along this direction. (this step will need 2 rotation operations)

→ then do the ' θ ' rotation around new z-axis.

→ Now rotate coordinate axes to old positions. (2 rotations in reverse again)

→ You will have to do 5 rotations total.

(Just an example only, i won't elaborate but

you can study from _____ . if need

help to understand. approach me directly,

Translation of origin in 2D, 3D

Till now all transformations resulted in

$$\text{Some } (x, y) \rightarrow (ax + by, cx + dz)$$

no extra const term. So we were writing matrix easily like

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

but what if we need to do a transformation

$$\text{like } (x, y) \rightarrow (a_1x + b_1y + c_1, a_2x + b_2y + c_2)$$

then we add an extra const dimension

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

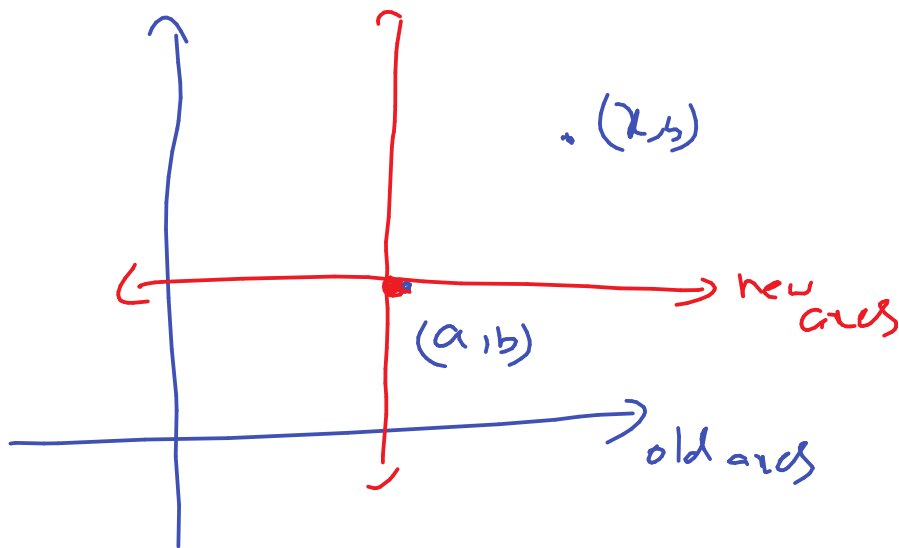
Shifting origin from $(0,0)$ to (a,b)

Here our axes we are moving to (a,b)

(or) Keep axes same place, grab XY plane by point (a,b) and drag it to origin.

So w.r.t new system every point

(x,y) becomes $(x-a, y-b)$



(x,y) in old system



Same Point appears
at $(x-a, y-b)$
in new system.

TM for this

$$(x,y) \rightarrow (x-a, y-b)$$

$$\begin{matrix} \Downarrow \\ \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -a \\ 0 & 1 & -b \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \end{matrix}$$

in 3D if you shift origin to (a, b, c)

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -a \\ 0 & 1 & 0 & -b \\ 0 & 0 & 1 & -c \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

if you want to combine any rotation, origin shift you can write rotation matrix as

$$\begin{bmatrix} C & -S & 0 & 0 \\ S & C & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{instead of} \quad \begin{bmatrix} C & -S & 0 \\ S & C & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Solve the practice questions for more understanding